

主成分分析

Principal Component Analysis

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声明

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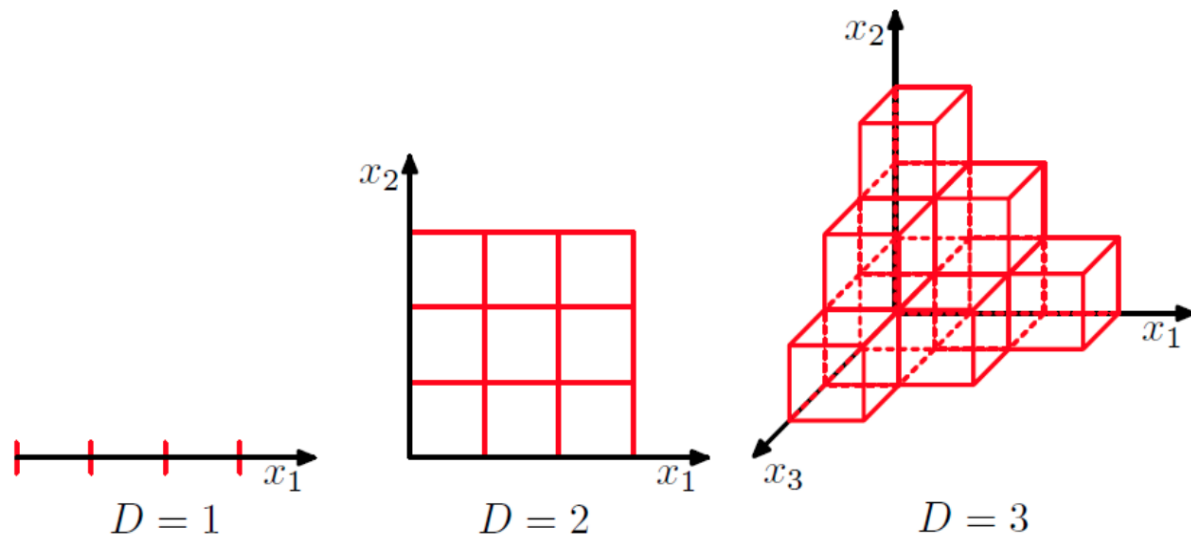
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动机

Motivation

- 从观念上讲，填充等量范围所需样本的数目，高纬度时比低纬度时要多很多，如下图所示：

From conception, the number of samples to fill up the same conceptual range of higher dimension will require exponentially large # of examples:



动机

Motivation

- ▶ 但由于采样成本的原因，对于观念上同样的区域，我们倾向于采样同样数量的样本。但由于对于高维度空间，我们可能需要更为复杂的模型来建模系统。但过拟合与采样，很多时候与样本数目有密切的关系。当模型更为复杂时，需要数目更多的样本。反过来说，当维度更少，过拟合的概率更低，泛化能力也更强。

However, due to sampling costs, we tend to sample the same number of examples for conceptually identical size of regions. But for high-dimensional spaces, we may need more complex models to modelling the system. But, overfitting and sampling are often closely related to the number of samples. More complex models require a larger number of samples. Conversely, fewer dimensions lead to a lower probability of overfitting and stronger generalization.

- ▶ 此时，降维是缓解维度问题的一种方法，也是我们发展主成分分析方法的动因。

Dimensionality reduction is a way to mitigate the curse of dimensionality, and this is also the motivation for developing the methods of principal component analysis.

数学背景

Math Background

- 称维度为 $D \times K$ 的矩阵 $U = [u_1, u_2, \dots, u_K]$ 为投影矩阵，其中 $D > K$ ， $u_j \in \mathbb{R}^D$ 称为投影方向向量。对于任意的 $x^{(i)} \in \mathbb{R}^D$ ，通过 U 将 $x^{(i)}$ 映为 $\mathbb{R}^K$ 中一点 $z^{(i)}$ ，即 $z^{(i)} = U^T x^{(i)}$ ，称其为 $x^{(i)}$ 的投影或低维嵌入。

A projection matrix is a matrix $U = [u_1, u_2, \dots, u_K]$ of size $D \times K$. Here $D > K$ and $u_j \in \mathbb{R}^D$ defines one projection direction. For $\forall x^{(i)} \in \mathbb{R}^D$, we can use U to project $x^{(i)}$ to $z^{(i)}$ in a lower dimension space $\mathbb{R}^K$, aka., $z^{(i)} = U^T x^{(i)}$. $z^{(i)}$ is also called low-dimensional embedding of $x^{(i)}$.

$$\begin{array}{c} K \times 1 \\ \text{orange bar} \\ z_i \end{array} = \begin{array}{c} K \times D \\ \text{gray box} \\ \begin{array}{c} \text{--- } u_1^T \text{ ---} \\ U \\ \text{--- } u_K^T \text{ ---} \end{array} \end{array} * \begin{array}{c} \text{blue bar} \\ x^{(i)} \end{array}$$

数学背景

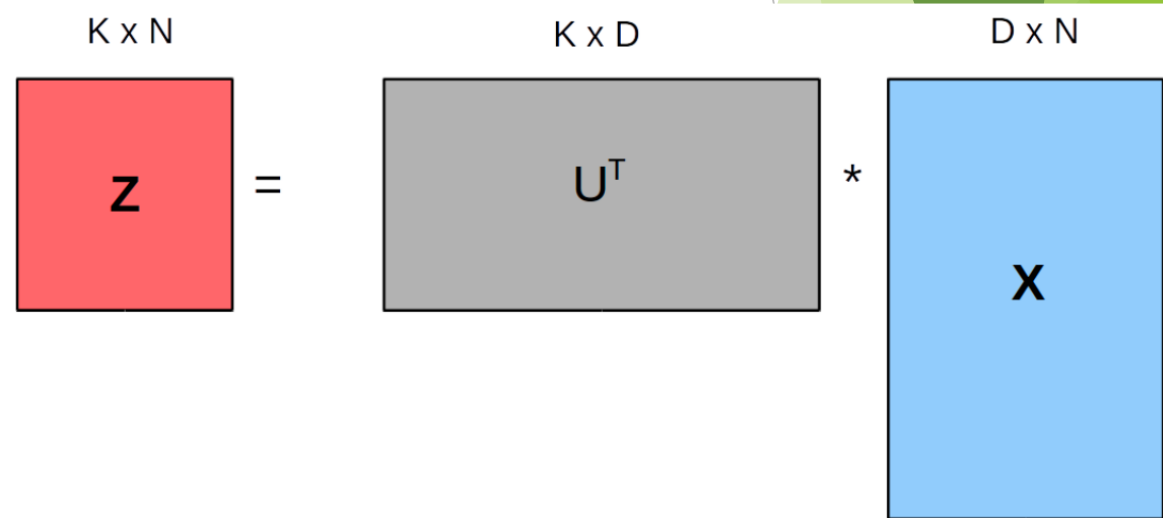
Math Background

- 将上述概念推广，即对具有 N 个数据点的集合，我们将其写成 $X = [x^{(1)}, x^{(2)}, \dots, x^{(N)}]$ ，其嵌入记为 $Z = [z^{(1)}, z^{(2)}, \dots, z^{(N)}]$ ，即 $Z = U^T X$ 。

We generalize the above concept. For a set with N data points, we will denote it as $X = [x^{(1)}, x^{(2)}, \dots, x^{(N)}]$, the corresponding embedding denoted as $Z = [z^{(1)}, z^{(2)}, \dots, z^{(N)}]$, aka., $Z = U^T X$.

- 上面只是从数学上描述，在实际问题中，我们该如何选取最优的投影矩阵 U ？在学习 U 时，我们应该优化什么准则？主成分分析是解决该方法。

The above is only a description from mathematical perspective. How do we choose the best projection matrix U ? What criteria should we optimize for when learning U Principle Component Analysis (PCA) is an algorithm for doing this.



概述

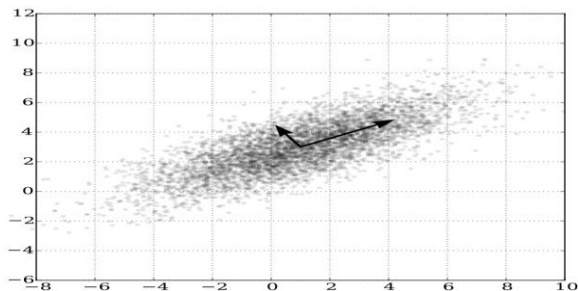
Summary

- ▶ 主成分分析是一种被广泛应用的技术，常用于降维、有损数据压缩、特征提取与数据可视化等场景。

它有两种刻画方式：

- ▶ 学习能捕获数据最大方差的投影方向
- ▶ 学习能最小化重构误差的投影方向

从数学的角度，PCA也可被理解为：改变数据表示的基，并对特征进行变换，使新特征彼此不相关。



- ▶ PCA is a technique widely used for applications such as dimensionality reduction, lossy data compression, feature extraction, and data visualization.

Two commonly used depictions:

- ▶ Learning projection directions that capture maximum variance in data
- ▶ Learning projection directions that result in smallest reconstruction error

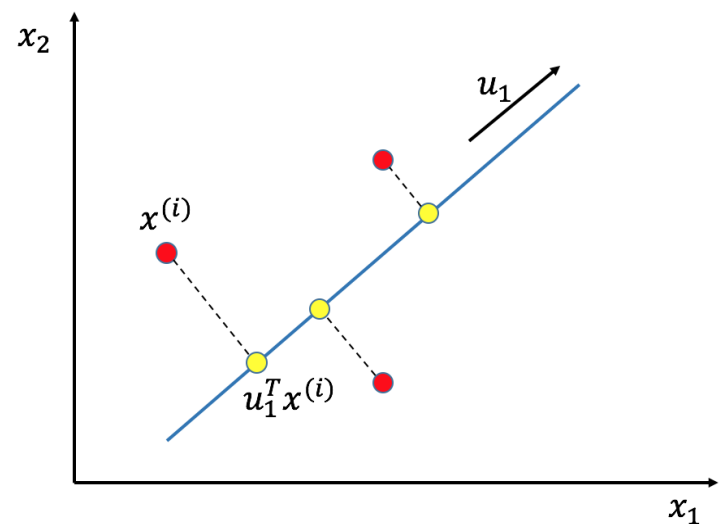
Can also be seen as changing the basis in which the data is represented (and transforming the features such that new features become decorrelated).

投影方差

Variance Captured by Projections

- 我们首先来算一下投影后的方差。考虑由 $u_1 \in \mathbb{R}^D$ 张成的一维子空间，为方便起见，令 $\|u_1\| = 1$ 。考虑任意的 $x^{(i)} \in \mathbb{R}^D$ ，其在该子空间中的投影为 $u_1^T x^{(i)}$ ，则所有投影的均值为 $\frac{1}{N} \sum_{i=1}^N u_1^T x^{(i)} = u_1^T \frac{1}{N} \sum_{i=1}^N x^{(i)}$ 。不难看出，所有 $x^{(i)}$ 的均值为 $\mu = \frac{1}{N} \sum_{i=1}^N x^{(i)}$ ，所以 $u_1^T \frac{1}{N} \sum_{i=1}^N x^{(i)} = u_1^T \mu$ 。

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投影方差

Variance Captured by Projections

- 此时，我们可以计算一下投影数据的方差：

Now we can calculate the variance of the projected data:

$$\frac{1}{N} \sum_{i=1}^N (u_1^T x^{(i)} - u_1^T \mu)^2 = \frac{1}{N} \sum_{i=1}^N [u_1^T (x^{(i)} - \mu)]^2 = u_1^T S u_1$$

这里 S 为协方差矩阵，维度为 $D \times D$ ，定义如下：

Here S is the $D \times D$ data covariance matrix:

$$S = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - \mu)(x^{(i)} - \mu)^T$$

- 注意，投影后数据的方差衡量了点的散布程度，若数据已经中心化（均值 $\mu = 0$ ），则协方差矩阵为 $S = \frac{1}{N} \sum_{i=1}^N x^{(i)}(x^{(i)})^T$ 。

Notably, variance measures the spreading/scattering extent of the projected data. If data already centred at $\mu = 0$, then $S = \frac{1}{N} \sum_{i=1}^N x^{(i)}(x^{(i)})^T$.

优化问题

Optimization Problem

- ▶ 直观地讲，特别对于分类问题，如果数据在输入空间已经具有很好的区分度，则建立模型处理要容易得多。因此，我们希望投影后的数据的方差最大。由于在方差的表达式 $u_1^T S u_1$ 中，只有 u_1 可变，因此，我们需要求解如下优化问题：

Intuitively speaking, especially for classification problem, if the data are quite separable in input space, it is much easier to setup models to process them. Hence, we hope the variance is as large as possible for projected data. Since in the expression of projected variance, namely $u_1^T S u_1$, only u_1 can be varying, therefore, we need to solve the following optimization problem:

$$\begin{aligned} & \max_{u_1} u_1^T S u_1 \\ & \text{s.t. } u_1^T u_1 = 1 \end{aligned}$$

- ▶ 我们用前面讲过的拉格朗日乘数法来求解上面的优化问题，其表达式如下：

We solve the optimization using Lagrange multipliers method, the expression is in the below:

$$\mathcal{L}(u_1, \lambda_1) = u_1^T S u_1 - \lambda_1 (u_1^T u_1 - 1)$$

这里 λ_1 为拉格朗日乘子。

Here λ_1 is a Lagrange multiplier.

优化问题

Optimization Problem

- 对 u_1 求导并令其为 0，可得 $Su_1 = \lambda_1 u_1$ ，因此， u_1 是矩阵 S 的特征向量（对应的特征值为 λ_1 ）。但它是 S 的哪一个特征向量呢？注意到由于 $u_1^T u_1 = 1$ ，投影后数据的方差为 $u_1^T S u_1 = \lambda_1$ 。当 u_1 取特征值最大的那个特征向量时，方差达到最大。这个最大特征值对应的特征向量 u_1 也被称为第一主成分。

Taking the derivative w.r.t. u_1 and setting to zero gives $Su_1 = \lambda_1 u_1$. Thus u_1 is an eigenvector of S (with corresponding eigenvalue λ_1). But which of S 's eigenvectors it is? Note that since $u_1^T u_1 = 1$, the variance of projected data is $u_1^T S u_1 = \lambda_1$. Therefore, var. is maximized when u_1 is the top eigenvector with largest eigenvalue. And the top eigenvector u_1 is also known as the first Principle Component.

- 同理，利用矩阵 S 的特征分解，可以求出其他方向（每个方向都与之前所有方向正交）——这就是主成分分析（PCA）。

Other directions can also be found likewise (with each being orthogonal to all previous ones) using the eigen-decomposition of S (this is PCA).

主成分分析步骤

Steps in PCA

- ▶ 中心化数据，即将每个数据点减去其平均 $\mu = \frac{1}{N} \sum_{i=1}^N x^{(i)}$;
 - ▶ 计算协方差矩阵 $S = \frac{1}{N} \sum_{i=1}^N x^{(i)} (x^{(i)})^T = \frac{1}{N} X X^T$;
 - ▶ 将协方差矩阵 S 进行特征值分解;
 - ▶ 将特征值从大到小排序，取最大的 K 个，即 $\{\lambda_l\}_{l=1,2,\dots,K}$ ，与对应的特征向量 $\{u_l\}_{l=1,2,\dots,K}$;
 - ▶ 最后，数据的嵌入由下面公式给出： $Z = U^T X$ ，其中 $U = [u_1, u_2, \dots, u_K] \in \mathbb{R}^{D \times K}$ 。
- ▶ Center the data, aka., subtract the mean $\mu = \frac{1}{N} \sum_{i=1}^N x^{(i)}$ from each data point;
 - ▶ Compute the covariance matrix $S = \frac{1}{N} \sum_{i=1}^N x^{(i)} (x^{(i)})^T = \frac{1}{N} X X^T$;
 - ▶ Do an eigen-decomposition of the covariance matrix S ;
 - ▶ Take the first K leading eigenvectors $\{\lambda_l\}_{l=1,2,\dots,K}$ with eigenvalues $\{u_l\}_{l=1,2,\dots,K}$;
 - ▶ The final K dim. projection of data is given by $Z = U^T X$ where U is $D \times K$.

重构误差

Reconstruction Error

- 下面，我们从重构误差的角度来求解PCA问题。考虑空间 $\mathbb{R}^D$ 的一组正交基 $u_1, u_2, \dots, u_D$ ，对任意的 $x^{(i)} \in \mathbb{R}^D$ ，我们可以用新的正交基将其表示为 $x^{(i)} = \sum_{l=1}^D z_l^{(i)} u_l$ ，具体地：

Now we try to interpret PCA from reconstruction error perspective. Consider a set of complete orthonormal basis vectors $u_1, u_2, \dots, u_D$ in $\mathbb{R}^D$, We can represent each data point $x^{(i)} \in \mathbb{R}^D$ exactly using the new basis: $x^{(i)} = \sum_{l=1}^D z_l^{(i)} u_l$. In detail:

$$\begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \\ \vdots \\ x_D^{(i)} \end{bmatrix} = [u_1 \quad u_2 \quad \cdots \quad u_D] * \begin{bmatrix} z_1^{(i)} \\ z_2^{(i)} \\ \vdots \\ z_D^{(i)} \end{bmatrix}$$

如果令 $z^{(i)} = [z_1^{(i)}, z_2^{(i)}, \dots, z_D^{(i)}]^T$ ， $U = [u_1, u_2, \dots, u_D]$ ，注意 $U^T U = 1$ ，我们有 $x^{(i)} = U z^{(i)}$ ， $z^{(i)} = U^T x^{(i)}$ 。注意此时，对 $z^{(i)}$ 的每一个元素 $z_l^{(i)}$ ，其为 $z_l^{(i)} = U^T x_l^{(i)}$ 。

If we denote $z^{(i)} = [z_1^{(i)}, z_2^{(i)}, \dots, z_D^{(i)}]^T$ ， $U = [u_1, u_2, \dots, u_D]$ ，and using $U^T U = 1$ ，we have $x^{(i)} = U z^{(i)}$ ， $z^{(i)} = U^T x^{(i)}$. Also note that each component of vector $z^{(i)}$ is $z_l^{(i)} = U^T x_l^{(i)}$.

重构误差

Reconstruction Error

- ▶ 显然，根据线性代数的知识，如果我们用所有的 D 个正交向量，则可以完全从 $z^{(i)}$ 重构 $x^{(i)}$ 。下面，我们考虑仅用 K ($K < D$) 个基向量逼近 $x^{(i)}$ ，即 $x^{(i)} \approx \sum_{l=1}^K z_l^{(i)} u_l$ 。如果令 $K = 1$ ，将 u_1 张成的空间视为投影空间，则 $x^{(i)}$ 的嵌入为 $z^{(i)} = u_1^T x^{(i)}$ 。现在，我们尝试从 $z^{(i)}$ 重构 $x^{(i)}$ ： $\tilde{x}^{(i)} = u_1 z^{(i)} = u_1 u_1^T x^{(i)}$ 。

Obviously, according to linear algebra, reconstruction of $x^{(i)}$ from $z^{(i)}$ will be exact if we use all D basis vectors. Now, we consider to approximate $x^{(i)}$ only use K ($K < D$) basis vectors, aka., $x^{(i)} \approx \sum_{l=1}^K z_l^{(i)} u_l$. Let's use $K = 1$ basis vector. Then, the one-dim embedding of $x^{(i)}$ into the subspace spanned by u_1 is $z^{(i)} = u_1^T x^{(i)}$. Now we try to reconstruct $x^{(i)}$ from its embedding $z^{(i)}$ as follows: $\tilde{x}^{(i)} = u_1 z^{(i)} = u_1 u_1^T x^{(i)}$.

- ▶ 则重构所有数据点时的总误差或总损失如下所示：

Then the total error or loss in reconstructing all the data points is as follows:

$$\ell(u_1) = \sum_{i=1}^N \|x^{(i)} - \tilde{x}^{(i)}\|^2 = \sum_{i=1}^N \|x^{(i)} - u_1 u_1^T x^{(i)}\|^2$$

重构误差

Reconstruction Error

- 我们希望找到可以最小化重构误差的 u_1 ，为此，我们将重构误差展开：

We want to find u_1 that minimize the reconstruction error, so, we rewrite the loss:

$$\begin{aligned}\ell(u_1) &= \sum_{i=1}^N \|x^{(i)} - u_1 u_1^T x^{(i)}\|^2 = \sum_{i=1}^N (x^{(i)} - u_1 u_1^T x^{(i)})^T (x^{(i)} - u_1 u_1^T x^{(i)}) \\&= \sum_{i=1}^N \left((x^{(i)})^T - (u_1 u_1^T x^{(i)})^T \right) (x^{(i)} - u_1 u_1^T x^{(i)}) \\&= \sum_{i=1}^N (x^{(i)})^T x^{(i)} - (x^{(i)})^T u_1 u_1^T x^{(i)} - (x^{(i)})^T u_1 u_1^T x^{(i)} + (x^{(i)})^T u_1 u_1^T u_1 u_1^T x^{(i)} \\&= \sum_{i=1}^N (x^{(i)})^T x^{(i)} - (x^{(i)})^T u_1 u_1^T x^{(i)} - (x^{(i)})^T u_1 u_1^T x^{(i)} + (x^{(i)})^T u_1 u_1^T x^{(i)} \\&= \sum_{i=1}^N (x^{(i)})^T x^{(i)} - (x^{(i)})^T u_1 u_1^T x^{(i)} = \sum_{i=1}^N (x^{(i)})^T x^{(i)} - u_1^T x^{(i)} (x^{(i)})^T u_1\end{aligned}$$

$$u_1^T u_1 = 1$$

重构误差

Reconstruction Error

- 从重构误差的表达式可以看出，最小化重构所有数据点的误差等价于求解下面优化问题：

From the expression of reconstruction error we can see that minimizing the error of reconstructing all the data points is equivalent to solving the following optimization problem:

$$\max_{u_1: \|u_1\|=1} u_1^T \left(\sum_{i=1}^N x^{(i)} (x^{(i)})^T \right) u_1 = \max_{u_1: \|u_1\|=1} u_1^T S u_1$$

假设数据已中心化，此时 S 为数据的协方差矩阵，这与我们最大化方差时所采用的目标函数是一致的，可以采用我们之前的方法求解。

Assume data is already centered, then S is the covariance matrix of the data. It is the same objective that we had when we maximized the variance, and can be solved with the solution we introduced before.